Module 7 Vocabulary

Angle Bisector: A line or segment that divides an angle into two equal parts.

Arc (in a construction): A portion of a circle drawn using a compass, typically used to locate points or measure distances in geometric constructions.

Arc Measure: The degree measure of an arc, which is equal to the measure of its corresponding central angle.

Center: The point inside a circle that is equidistant from all points on the circle.

Chord: A line segment with both endpoints on the circle.

Central Angle: An angle whose vertex is the center of the circle and whose sides are radii.

Circle: The set of all points in a plane that are equidistant from a given point called the center.

Circumcenter: The point where the **perpendicular bisectors** of a triangle's sides intersect; it is the center of the **circumscribed circle**.

Circumscribed Circle: A circle that passes through all three vertices of a triangle, with its center at the circumcenter.

Compass: A geometric tool used to draw circles and arcs by maintaining a fixed radius.

Concurrent Lines: Three or more lines that intersect at a single point.

Cyclic Quadrilateral: A quadrilateral whose vertices all lie on the circle.

Diameter: A chord that passes through the center of the circle, making it the longest chord.

Exterior Angle: An angle formed outside the circle by two secants, two tangents, or a secant and a tangent.

External Segment: The portion of a secant segment that lies outside the circle, extending from an external point to the circle's boundary.

Hexagon: A six-sided polygon.

Incenter: The point where the **angle bisectors** of a triangle intersect; it is the center of the **inscribed circle**.

Inscribed Angle: An angle whose vertex is on the circle and whose sides are chords of the circle.

Inscribed Circle: A circle that fits inside a triangle, touching all three sides at exactly one point each, with its center at the **incenter**.

Inscribed Hexagon: A hexagon where all six vertices lie on a circle.

Inscribed Polygon: A polygon where all of its vertices lie on a circle.

Inscribed Quadrilateral: A quadrilateral whose vertices all lie on a circle, also called a **cyclic quadrilateral**.

Inscribed Triangle: A triangle with all three vertices on a circle.

Intercepted Arc: The arc between two points where an angle intersects the circle.

Internal Segment: The portion of a secant segment that lies inside the circle, extending from the point where the secant enters the circle to the second intersection point.

Major Arc: The part of a circle's circumference that is larger than a semicircle.

Minor Arc: The part of a circle's circumference that is smaller than a semicircle.

Opposite Angles: Two angles in a quadrilateral that are not adjacent to each other.

Orthocenter: The point where the **altitudes** of a triangle intersect.

Perpendicular: Two lines or segments that intersect at a 90° angle.

Perpendicular Bisector: A line that divides a segment into two equal parts at a 90-degree angle.

Point of Tangency: The point at which a tangent line touches the circle.

Radius: A line segment from the center of the circle to any point on the circle.

Secant: A line that intersects a circle at two points.

Segment: A part of a line that has two endpoints and a measurable length. In a circle, segments can include chords, secant segments, radii, and diameters.

Semicircle: Half of a circle, formed by a diameter dividing the circle into two equal arcs.

Straightedge: A tool used to draw straight lines, typically without measurement markings.

Tangent: A line that touches a circle at exactly one point without crossing it.

Triangle: A three-sided polygon with three angles that sum to **180°**.